MAR 657 Visual Analytics
Assignment 4: Group Final Project Proposal

Team Member 1: Atharv Prashant Tungatkar U01984472

Team Member 2: Pace ID

Team Member 3: Pace ID

Team Member 4:

OBJECTIVE/ PROBLEM:

Project Title: Factors influencing Housing prices in NYC

Problem Statement:

The New York City real estate market is one of the globe's most complex and dynamic. While property value per se—location, property size, amenities—is all that must be taken into account for many markets, realizing neighborhood-level considerations of safety, environmental status, and housing complaints can enable one to find the role in which livability and public health indicators influence real estate price and rental affordability.

This study combines a number of datasets from NYC Open Data and other public datasets to conduct a data-driven analysis of the key drivers of residential property value. In bridging economic, social, and environmental factors, the project aims to provide an end-to-end view of urban housing trends within boroughs.

Primary Research Questions:

1) Which neighborhood-level factors—such as crime rate, air quality, and housing complaints—most significantly influence residential property prices across NYC?

2) Are these factors correlated or interdependent (e.g., do low air quality and high complaint density coexist with lower property values)?
3) Can we build a predictive model using these factors to estimate future housing price trends or affordability levels?

Supporting Research questions:

1) How do property price drivers differ across boroughs (Manhattan vs. Bronx vs. Queens)?
2) Does the presence of nearby high-performing schools mitigate the negative impact of crime on property value?
3) Are areas with poor air quality and higher pest complaints experiencing slower property appreciation?
4) How does housing affordability correlate with neighborhood livability and infrastructure quality?

Justification:

This effort is of immense significance to diverse stakeholders:

Real estate developers and investors: Identify undervalued locations with their social and environmental condition enhancing.

Urban planners and policymakers: Prioritize investments in infrastructure, education, and sanitation where the effect on property stability and affordability would be optimal.

Banks and financial institutions: Assess real estate risk and investment prospect in terms of non-traditional urban metrics.

Household residents: Understand how neighborhood-level information can sustain price fluctuations and identify improved deals.

Through the integration of a number of urban datasets, the project provides evidence-based information to inform equitable development of housing and sustainable urban planning.

Data sources:

1) NYC Department of Finance — Property Rolling Sales Data

   https://data.cityofnewyork.us/City-Government/NYC-Citywide-Annualized-Calendar-Sales-Update/w2pb-icbu/about_data

   Description: The dataset contains housing sales prices in different boroughs from 2018-2025.

   Annualized sales file displays yearly sales information of properties sold in New York City.

2) NYPD Complaint Data (Historic)

https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Historic/

Description: This dataset includes all valid felony, misdemeanor, and violation crimes reported to the New York City Police Department (NYPD) from 2006 to the end of last year (2019). For additional details, please see the attached data dictionary in the 'About' section.

3) NYC Rodent Inspection

https://data.cityofnewyork.us/Health/Rodent-Inspection/p937-wjvj/about_data

Description: Dataset contains information on rat inspections.
Rat Information Portal Data Release Notes April 20, 2015
The Rat Information Portal (RIP) is a web-based mapping application where users can view rat inspection data.

4) NYC 311 Service Request

https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2010-to-Present/erm2-nwe9/data_preview

Description: All 311 Service Requests from 2010 to present. This information is automatically updated daily.

5) NYC Air Quality Data (DOHMH)

https://data.cityofnewyork.us/Environment/Air-Quality/c3uy-2p5r/about_data

Description:

Dataset contains information on New York City air quality surveillance data.

Approach/Methodology:

The whole project will be conducted in R Studio, with emphasis on visualization and geospatial analysis as well as statistical and predictive modeling. Cleaning and merging multiple datasets from NYC Open Data, including property transactions, crime reports, school performance, 311 complaints, rodent inspections, and air quality readings, is the first step. With R packages such as RSocrata, readr, tidycensus, dplyr, and sf, spatial identifiers such as ZIP codes, tracts, and coordinates will be normalized to merge datasets into an integrated master dataset. New variables such as affordability ratios and complaint densities per 1,000 residents will be engineered to introduce additional analytical dimensions.

The second step is reserved for exploratory data analysis and visualization. Patterns of crime within space-time, air contamination, housing-related grievances, and pest infestation will be mapped using tmap and leaflet to identify hotspots and areas of interest between boroughs. Choropleth maps, scatter plots, density plots, and bivariate maps will be used to explore correlations between variables, e.g., the correlation between quality of schools and house price or air quality and affordability. Correlation heatmaps and summary statistics will allow interdependencies to be revealed and guide more detailed analysis. Visualization will be the primary lens that findings are taken through, allowing patterns to be easily explained to both technical and non-technical readers.

Analytical modeling will use hedonic price regression models like OLS and Ridge regression to estimate the influence of neighborhood-level determinants on property value, controlling for property type like building size, age, and type. Spatial regression techniques will be employed to account for neighborhood spillover effects. Predictive models using Random Forest or XGBoost will also be developed to forecast property prices and assess the relative importance of various factors. Visualization of model results such as feature importance plots and residual maps will aid in interpreting results and pinpointing areas of under- or over-prediction.

Finally, a dynamic interactive Shiny dashboard will bring together the visualizations to enable dynamic data exploration. They will be able to search by neighborhood or borough, view choropleth maps of crime, air quality, and pest complaint maps, contrast school quality against property price, and view temporal trends in property value and environmental conditions between 2018 and 2024. Time-series plots, interactive scatter plots, and heatmaps will provide a multi-dimensional perspective, making the insights intuitive and actionable for developers, policymakers, investors, and prospective homebuyers. The approach ensures that visualization remains at the center of the project, with analyses being designed to express spatial and temporal trends in a simple and informative manner.

Anticipated Conclusions:

- Crime rates, housing complaints, and pest infestations negatively influence property prices.
- Better school quality and air quality positively correlate with higher housing values.
- Neighborhoods with improving livability indicators show greater appreciation potential.
- Affordability challenges are concentrated in areas with high livability but limited housing supply.

Project Timelines:

• Oct 7 – Submit Proposal
• Oct 9–12 – Data acquisition and schema setup
• Oct 13–17 – Cleaning, geocoding, and NTA joins
• Oct 18–22 – Build visuals (choropleths, heatmaps)
• Oct 23–26 – Regression and analysis
• Oct 27–30 – Draft poster and paper
• Oct 31–Nov 3 – Final polish and submission